

ECO6067 Empirical Replication Project

Card & Krueger (1994): Minimum Wages and Employment

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Abstract

This report presents an end-to-end replication and extension of Card and Krueger's (1994) study of the 1992 New Jersey minimum-wage increase. The original paper exploits a natural policy contrast: New Jersey raised its minimum wage from \$4.25 to \$5.05 per hour on 1 April 1992, while neighboring Pennsylvania did not. Card and Krueger surveyed fast-food restaurants in New Jersey and eastern Pennsylvania before and after the reform and compared changes in employment, wages and related outcomes.

The project rebuilds the analysis from the course-provided raw file `public.csv`, documents sample construction and variable definitions, reproduces the allocated core outputs (Tables 2-4 and Figure 1), validates the main numerical results against the published paper, and adds an independent extension on heterogeneity across restaurant chains. The raw file contains 410 restaurants and 46 original variables; the processed data retain all 410 rows and add constructed variables used by the replication.

The central replication result is a difference-in-differences estimate of approximately 2.75 full-time-equivalent (FTE) employees per store. This closely matches the published value of 2.76. Regression-adjusted specifications also reproduce the paper closely: the New Jersey coefficients in the first two Table 4 specifications are 2.326 and 2.304, compared with 2.33 and 2.30 in the paper, while the first two GAP coefficients are 15.653 and 14.916, compared with 15.65 and 14.92. The automated validation routine passes all 13 pre-specified checks.

The independent extension asks whether the relative employment response differs across Burger King, KFC, Roy Rogers and Wendy's. The joint heterogeneity test yields $F = 2.602$ and $p = 0.0518$. Thus equal chain-specific responses are not rejected at the 5% level, although the result is suggestive at the 10% level. Because several Pennsylvania chain subsamples are small, the extension is interpreted cautiously rather than as decisive evidence of heterogeneous effects.

1. Research Question, Economic Motivation and Original Study

1.1 Research question

The main replication question is:

Did New Jersey's 1992 minimum-wage increase reduce employment in the fast-food industry relative to comparable restaurants in eastern Pennsylvania?

Main hypothesis. Let δ denote the difference-in-differences coefficient defined in Section 3.1. The benchmark null is $H_0: \delta = 0$, while the directional alternative implied by the standard competitive labor-market model is $H_1: \delta < 0$. Thus, a negative value would be consistent with the conventional prediction that the higher minimum wage reduced New Jersey employment relative to Pennsylvania; a zero or positive estimate would not support that directional prediction. The regression tables retain conventional two-sided inference for the null of no differential effect.

The independent extension asks:

Did the employment response to New Jersey's minimum-wage increase differ across restaurant chains?

1.2 Economic motivation

In a standard competitive labor-market model, a binding increase in the minimum wage raises the cost of low-wage labor and may reduce employment. Card and Krueger (1994) study an unusually transparent policy setting in which one state changed its statutory minimum wage while an adjacent comparison area did not. Their design is therefore attractive for empirical evaluation because it compares changes in outcomes rather than simply comparing employment levels across places.

The empirical contrast is between New Jersey (treatment state) and eastern Pennsylvania (comparison state). The first survey was conducted before the policy change and the second after it. The paper’s headline conclusion is that it found no indication that New Jersey’s minimum-wage increase reduced employment in the sampled fast-food restaurants. The purpose of this project is not merely to restate that conclusion, but to reconstruct the analysis from the underlying data, verify the reported outputs, explain any discrepancies, and demonstrate a reproducible workflow.

1.3 Why this paper is suitable for replication

The study is well suited to a replication exercise for four reasons. First, the economic treatment is clearly defined and externally dated. Second, the main estimands can be expressed through transparent before-after and cross-state comparisons. Third, the course-provided replication data include the restaurant-level variables needed to reconstruct the paper’s main employment and wage measures. Fourth, the paper reports enough numerical detail to permit explicit validation rather than relying only on qualitative similarity.

2. Data Source, Sample Construction and Variable Definitions

2.1 Data source and repository structure

The project uses the course-provided `public.csv` file corresponding to the Card-Krueger fast-food survey. For reproducibility, the workflow expects the raw input at:

```
data/raw/public.csv
```

Data access and redistribution. The raw `public.csv` file is course-provided material and is not included in the public GitHub repository. An authorised user should obtain the course-provided dataset and place it locally at `data/raw/public.csv` using the exact filename. The remaining workflow then runs without any code changes. This follows the course instruction to exclude the CSV file from the public repository while documenting the required input location clearly.

The raw dataset contains **410 restaurants and 46 original variables**. The state indicator contains 331 New Jersey restaurants and 79 Pennsylvania restaurants. Chain codes correspond to Burger King, KFC, Roy Rogers and Wendy’s. The code never relies on machine-specific absolute paths; all files are addressed relative to the repository root.

The processed analysis file generated by the workflow is:

```
data/processed/analysis_sample.csv
```

It contains 410 rows and 56 columns after construction of analysis variables. This processed row-level file is generated locally by the workflow. The public GitHub repository stores the code, documentation, generated summary tables and generated figures under `outputs/`, while the course-provided raw CSV is not redistributed.

2.1.1 Raw variables used in the replication

The original data dictionary describes `public.csv` as store-level survey data with 410 observations and 46 columns. The variables below are the raw fields used most directly in the replication. A suffix “2” denotes the second survey; blank entries in the raw file denote missing values.

Raw variable(s)	Definition in the supplied data dictionary	Role in this replication
SHEET	Original store/survey sheet identifier.	Record identification and raw-data checking.
CHAIN	Restaurant chain: 1 = Burger King; 2 = KFC; 3 = Roy Rogers; 4 = Wendy’s.	Chain controls and the chain-heterogeneity extension.
CO_OWNED	Company ownership indicator: 1 = company-owned.	Ownership control in regression-adjusted specifications.

Raw variable(s)	Definition in the supplied data dictionary	Role in this replication
STATE	State indicator: 1 = New Jersey; 0 = Pennsylvania.	Defines the treatment and comparison groups.
SOUTHJ, CENTRALJ	Indicators for stores in southern and central New Jersey.	Regional controls in the most saturated Table 4 specification.
PA1, PA2	Pennsylvania-area indicators: northeastern Philadelphia suburbs and the Easton area.	Regional controls for Pennsylvania stores.
EMPFT, EMPFT2	Number of full-time employees in the first and second surveys.	Inputs to pre- and post-policy FTE employment.
EMPPT, EMPPT2	Number of part-time employees in the first and second surveys.	Inputs to FTE employment, with each part-time worker weighted by 0.5.
NMGRS, NMGRS2	Number of managers and assistant managers in the first and second surveys.	Inputs to pre- and post-policy FTE employment.
WAGE_ST, WAGE_ST2	Starting wage, in dollars per hour, in the first and second surveys.	Used in Table 2, Figure 1, and construction of the initial wage-gap variable.
BONUS	Indicator equal to 1 if a cash recruiting bonus is offered for new workers.	First-wave labor-practice descriptor used in descriptive replication.
SPECIAL2	Indicator equal to 1 if a special program is offered for new workers in the second survey.	Second-wave labor-practice descriptor.
HRSOPEN, HRSOPEN2	Hours open per day in the first and second surveys.	Operational-hours measure used in descriptive replication.
PSODA, PSODA2	Price of a medium soda, including tax, in the first and second surveys.	Menu-price measure used in descriptive replication.
PFRY, PFRY2	Price of small fries, including tax, in the first and second surveys.	Menu-price measure used in descriptive replication.
PENTREE, PENTREE2	Price of an entrée, including tax, in the first and second surveys.	Menu-price measure used in descriptive replication.
STATUS2	Second-wave status: 0 = refused; 1 = answered; 2 = renovation closure; 3 = permanent closure; 4 = highway-construction closure; 5 = mall-fire closure.	Used to identify second-wave response status and closure-related sensitivity calculations.

These raw fields are transformed into the analysis variables described below. In particular, EMPFT/EMPPT/NMGRS and their second-wave counterparts generate FTE employment, STATE defines the New Jersey treatment group, WAGE_ST enters the wage-gap construction, and CHAIN identifies the subgroups used in the independent extension.

2.2 Full-time-equivalent employment

Following the paper, full-time-equivalent employment counts each part-time worker as half of a full-time worker. With the variables used in the data, the measure is

$$FTE_i = EMPFT_i + NMGRS_i + 0.5 EMPPT_i.$$

The corresponding second-wave measure is constructed from the second-wave employment variables. The change in employment used in the first-difference regressions is

$$\Delta FTE_i = FTE_{i,2} - FTE_{i,1}.$$

This definition is central to Tables 2-4. The raw data inspection identified missing components in the employment variables, so the scripts preserve the paper's use of available observations for descriptive means and then apply the appropriate balanced or complete-data restrictions for specific models.

2.3 Treatment, timing and wage-gap variables

Let $NJ_i = 1$ for a restaurant in New Jersey and $NJ_i = 0$ for a restaurant in Pennsylvania. For the two-wave representation, let $POST_t = 1$ after the policy change and $POST_t = 0$ before it.

The paper also uses an initial wage-gap measure, which captures how large a proportional wage increase would have been required for an affected New Jersey store to reach the new minimum wage. The replication implements the concept as

$$GAP_i = NJ_i \max \left\{ 0, \frac{5.05 - W_{i,1}}{W_{i,1}} \right\},$$

where $W_{i,1}$ is the starting wage in wave 1. Thus Pennsylvania stores have $GAP_i = 0$, while New Jersey stores already paying at least \$5.05 also have zero gap.

2.4 Sample construction by analysis stage

The project deliberately does not force every stage onto one common complete-case sample, because the paper uses different samples for different outputs.

- **Table 2** uses available nonmissing observations for each reported variable.
- **Table 3** reports all-available means, a balanced employment sample, and a sensitivity calculation that treats temporary closures as zero employment.
- **Table 4** uses the paper's restricted estimation sample with employment and starting-wage information in both waves; the replication sample contains **357 stores**, matching the paper.
- **Independent extension** uses a balanced employment sample with nonmissing FTE before and after, producing **384 restaurants**.

This distinction matters: forcing all results to the strictest complete-case sample would make the workflow simpler, but would no longer reproduce the paper's design faithfully.

3. Empirical Design

3.1 Difference-in-differences logic

The core two-group, two-period difference-in-differences (DiD) estimand is

$$\widehat{DiD} = (FTE_{NJ,post} - FTE_{NJ,pre}) - (FTE_{PA,post} - FTE_{PA,pre}).$$

Equivalently, a two-period regression can be written as

$$FTE_{it} = \alpha + \beta NJ_i + \gamma POST_t + \delta(NJ_i \times POST_t) + \varepsilon_{it},$$

where δ is the DiD coefficient. In the first-difference representation used in the paper's Table 4, the same cross-state comparison is represented by a regression of ΔFTE_i on the New Jersey indicator and additional controls.

The causal interpretation requires Pennsylvania to provide a credible counterfactual for the employment change that New Jersey stores would have experienced without the policy. With only one pre-policy survey wave, a conventional multi-period parallel-trends test is not available. Card and Krueger therefore rely on institutional comparability and additional specification checks, including comparisons within New Jersey between initially low- and high-wage stores.

3.2 Regression-adjusted models

The paper estimates reduced-form models of the form

$$\Delta FTE_i = \alpha + \beta NJ_i + X_i' \gamma + \varepsilon_i,$$

and

$$\Delta FTE_i = \alpha + \theta GAP_i + X_i' \gamma + \varepsilon_i.$$

The control vector X_i is expanded across specifications to include restaurant-chain and ownership indicators, and finally regional indicators. These models test whether the simple DiD result is sensitive to observable composition and regional demand controls.

3.3 Validation strategy

Replication is evaluated against published benchmark values rather than by demanding exact equality at every decimal place. The validation script checks 13 core quantities with pre-specified tolerances. Small differences caused by rounding or implementation details are recorded rather than hidden. This approach is consistent with the assignment guidance that a defensible replication need not match every published number exactly if differences are documented and explained.

4. Core Replication Results

4.1 Table 2: descriptive statistics and variable reconstruction

The complete Table 2 replication is reported below rather than limiting the report to a small set of selected rows. For each variable, the table reports the New Jersey and Pennsylvania replication means, replication standard errors, the replication t-statistic, the state-specific sample counts, and the corresponding published benchmarks. Replication entries retain the precision of outputs/tables/table2_replication.csv, while the paper benchmarks retain the decimal precision shown in the original published table; no zeros are added simply to standardise the number of decimal places.

Table 1. Full replication of Card and Krueger Table 2

Panel A. Store types and ownership

Variable	NJ replication	NJ SE	PA replication	PA SE	Replication t	N NJ	N PA	Paper benchmark (NJ / PA / t)
Burger King (%)	41.087613	2.708331	44.303797	5.624527	-0.515197	331	79	41.1 / 44.3 / -0.5
KFC (%)	20.543807	2.224064	15.189873	4.063999	1.155666	331	79	20.5 / 15.2 / 1.2
Roy Rogers (%)	24.773414	2.376411	21.518987	4.653138	0.622875	331	79	24.8 / 21.5 / 0.6
Wendy's (%)	13.595166	1.886706	18.987342	4.440800	-1.117556	331	79	13.6 / 19.0 / -1.1
Company-owned (%)	34.138973	2.610252	35.443038	5.416138	-0.216899	331	79	34.1 / 35.4 / -0.2

Panel B. Wave 1

Variable	NJ replication	NJ SE	PA replication	PA SE	Replication t	N NJ	N PA	Paper benchmark (NJ / PA / t)
FTE employment	20.439408	0.508261	23.331169	1.351149	-2.003183	321	77	20.4 / 23.3 / -2.0
Full-time employees (%)	32.848870	1.325610	35.037798	2.727447	-0.721817	321	77	32.8 / 35.0 / -0.7
Starting wage	4.612134	0.019546	4.630132	0.040341	-0.401496	314	76	4.61 / 4.63 / -0.4
Wage = \$4.25 (%)	30.513595	2.534776	32.911392	5.320473	-0.406859	331	79	30.5 / 32.9 / -0.4
Price of full meal	3.351061	0.036544	3.042368	0.068950	3.955798	311	76	3.35 / 3.04 / 4.0
Hours open	14.418429	0.152770	14.525316	0.332061	-0.292428	331	79	14.4 / 14.5 / -0.3
Recruiting bonus (%)	23.564955	2.336268	29.113924	5.143794	-0.982207	331	79	23.6 / 29.1 / -1.0

Panel C. Wave 2

Variable	NJ replication	NJ SE	PA replication	PA SE	Replication t	N NJ	N PA	Paper benchmark (NJ / PA / t)
FTE employment	21.027429	0.520309	21.165584	0.943221	-0.128252	319	77	21.0 / 21.2 / -0.2
Full-time employees (%)	35.865552	1.379993	30.384849	2.806073	1.752677	314	76	35.9 / 30.4 / 1.8
Starting wage	5.080849	0.005863	4.617465	0.042425	10.819657	318	71	5.08 / 4.62 / 10.8
Wage = \$4.25 (%)	0.000000	0.000000	25.316456	4.923417	-5.142050	331	79	0.0 / 25.3 / —
Wage = \$5.05 (%)	85.196375	1.954959	1.265823	1.265823	36.037393	331	79	85.2 / 1.3 / 36.1
Price of full meal	3.414754	0.036397	3.026620	0.067206	5.078350	305	71	3.41 / 3.03 / 5.0
Hours open	14.419782	0.151888	14.653846	0.326984	-0.649206	321	78	14.4 / 14.7 / -0.8
Recruiting bonus (%)	20.317460	2.270657	23.376623	4.854726	-0.570793	315	77	20.3 / 23.4 / -0.6

The full output confirms the same descriptive pattern shown by the published table. In wave 1, FTE employment is 20.439408 in New Jersey and 23.331169 in Pennsylvania, compared with paper values of 20.4 and 23.3. Starting wages are 4.612134 and 4.630132, respectively. In wave 2, FTE employment is 21.027429 in New Jersey and 21.165584 in Pennsylvania, while starting wages are 5.080849 and 4.617465. The post-policy wage distribution also changes sharply: the Table 2 indicator for a \$5.05 starting wage is 85.196375% in New Jersey and 1.265823% in Pennsylvania, compared with published values of 85.2% and 1.3%. The close agreement is visible across store-type shares, employment, wages, menu prices, hours and recruiting-bonus variables, not only in the four headline quantities used in the earlier draft.

4.2 Table 3: employment levels, changes and the core DiD estimate

Table 3 is reproduced in full across the five output rows used by the workflow: employment before, employment after, the change in mean employment, the balanced-sample change, and the temporary-closure sensitivity calculation. Each panel reports the replication mean, the paper benchmark, their recorded difference, the replication and paper standard errors when present in the generated file, and the sample count when present. Replication entries retain the precision of the generated CSV, while paper values retain the precision displayed in the original Table 3. A dash is retained wherever the generated CSV contains no value; no standard error or sample size is invented.

Table 2. Full replication of Card and Krueger Table 3

Panel 1. FTE employment before, all available observations

Column	Replication mean	Paper mean	Mean difference	Replication SE	Paper SE	N
PA	23.331169	23.33	0.001169	1.351149	1.35	77
NJ	20.439408	20.44	-0.000592	0.508261	0.51	321
NJ - PA	-2.891761	-2.89	-0.001761	1.443583	1.44	—
NJ Low	19.556701	19.56	-0.003299	0.774856	0.77	97
NJ Mid	20.080657	20.08	0.000657	0.841274	0.84	137
NJ High	22.250000	22.25	0.000000	1.135799	1.14	71
Low - High	-2.693299	-2.69	-0.003299	1.374933	1.37	—
Mid - High	-2.169343	-2.17	0.000657	1.413429	1.41	—

Panel 2. FTE employment after, all available observations

Column	Replication mean	Paper mean	Mean difference	Replication SE	Paper SE	N
PA	21.165584	21.17	-0.004416	0.943221	0.94	77
NJ	21.027429	21.03	-0.002571	0.520309	0.52	319
NJ - PA	-0.138155	-0.14	0.001845	1.077213	1.07	—
NJ Low	20.877551	20.88	-0.002449	1.005902	1.01	98
NJ Mid	20.955556	20.96	-0.004444	0.755608	0.76	135
NJ High	20.213768	20.21	0.003768	1.025104	1.03	69
Low - High	0.663783	0.67	-0.006217	1.436202	1.44	—

Column	Replication mean	Paper mean	Mean difference	Replication SE	Paper SE	N
Mid - High	0.741787	0.75	-0.008213	1.273492	1.27	—

Panel 3. Change in mean FTE employment

Column	Replication mean	Paper mean	Mean difference	Replication SE	Paper SE	N
PA	-2.165584	-2.16	-0.005584	—	1.25	—
NJ	0.588021	0.59	-0.001979	—	0.54	—
NJ - PA	2.753606	2.76	-0.006394	—	1.36	—
NJ Low	1.320850	1.32	0.000850	—	0.95	—
NJ Mid	0.874899	0.87	0.004899	—	0.84	—
NJ High	-2.036232	-2.04	0.003768	—	1.14	—
Low - High	3.357082	3.36	-0.002918	—	1.48	—
Mid - High	2.911131	2.91	0.001131	—	1.41	—

Panel 4. Change in mean FTE employment, balanced sample

Column	Replication mean	Paper mean	Mean difference	Replication SE	Paper SE	N
PA	-2.283333	-2.28	-0.003333	1.253269	1.25	75
NJ	0.466667	0.47	-0.003333	0.480829	0.48	309
NJ - PA	2.750000	2.75	0.000000	1.342341	1.34	—
NJ Low	1.204787	1.21	-0.005213	0.822490	0.82	94
NJ Mid	0.709848	0.71	-0.000152	0.686227	0.69	132
NJ High	-2.156716	-2.16	0.003284	1.007321	1.01	67
Low - High	3.361504	3.36	0.001504	1.300455	1.30	—
Mid - High	2.866565	2.87	-0.003435	1.218853	1.22	—

Panel 5. Change in mean FTE employment, temporary closures = 0

Column	Replication mean	Paper mean	Mean difference	Replication SE	Paper SE	N
PA	-2.283333	-2.28	-0.003333	1.253269	1.25	75
NJ	0.225879	0.23	-0.004121	0.490979	0.49	313
NJ - PA	2.509212	2.51	-0.000788	1.346010	1.35	—
NJ Low	0.897368	0.90	-0.002632	0.869916	0.87	95
NJ Mid	0.494030	0.49	0.004030	0.692838	0.69	134
NJ High	-2.389706	-2.39	0.000294	1.019380	1.02	68
Low - High	3.287074	3.29	-0.002926	1.340108	1.34	—
Mid - High	2.883736	2.88	0.003736	1.232542	1.23	—

The core relative employment change is positive in every principal construction. Using all available observations, the generated output records a New Jersey mean change of 0.588021, a Pennsylvania mean change of -2.165584, and an NJ - PA difference of 2.753606, compared with the published 2.76. On the balanced employment sample the replicated NJ - PA change is exactly 2.750000, matching the 2.75 benchmark. When temporary closures are set to zero, the replicated NJ - PA change is 2.509212, compared with 2.51 in the paper. These results show that the positive relative change does not depend on a single treatment of missing follow-up employment.

The wage-group comparisons are also preserved. For the all-available change row, New Jersey low-wage stores have a mean change of 1.320850, mid-wage stores 0.874899, and high-wage stores -2.036232. The corresponding Low - High and Mid - High differences are 3.357082 and 2.911131. In the balanced sample they are 3.361504 and 2.866565, and in the temporary-closure sensitivity calculation they are 3.287074 and 2.883736. Reporting these rows makes the replication of the original Table 3 substantially more complete than a single headline DiD number.

4.3 Table 4: regression-adjusted employment models

The Table 4 replication uses 357 restaurants in every specification. The generated run reports a mean employment change of -0.238 and a standard deviation of 8.825 for this estimation sample. To keep the full output readable in the existing report layout, the coefficient and standard-error results are shown first, followed by the control and

model-diagnostic fields from the same CSV output. Replication entries retain the generated precision, while published values use the decimal precision shown in Card and Krueger's Table 4.

Table 3. Replication of Card and Krueger Table 4

Panel A. Coefficients and standard errors

Model	Main variable	Replication coef.	Paper coef.	Coef. difference	Replication SE	Paper SE	SE difference
(i)	STATE	2.325831	2.33	-0.004169	1.191596	1.19	0.001596
(ii)	STATE	2.303860	2.30	0.003860	1.195532	1.20	-0.004468
(iii)	GAP	15.652890	15.65	0.002890	6.080237	6.08	0.000237
(iv)	GAP	14.915674	14.92	-0.004326	6.205335	6.21	-0.004665
(v)	GAP	11.979237	11.91	0.069237	7.419121	7.39	0.029121

Panel B. Controls and model diagnostics

Model	Chain/ownership controls	Region controls	Replication SER	Paper SER	Replication control p	Paper control p	N
(i)	No	No	8.790861	8.79	—	—	357
(ii)	Yes	No	8.784658	8.78	0.344251	0.34	357
(iii)	No	No	8.756546	8.76	—	—	357
(iv)	Yes	No	8.759209	8.76	0.437360	0.44	357
(v)	Yes	Yes	8.752489	8.75	0.404570	0.40	357

Models (i)-(iv) reproduce the published coefficients and standard errors extremely closely. Model (i) gives a STATE coefficient of 2.325831 with standard error 1.191596, versus 2.33 and 1.19 in the paper. Model (ii) gives 2.303860 with standard error 1.195532, versus 2.30 and 1.20. For the wage-gap specifications, model (iii) gives 15.652890 with standard error 6.080237, while model (iv) gives 14.915674 with standard error 6.205335. Their published benchmarks are 15.65/6.08 and 14.92/6.21, respectively.

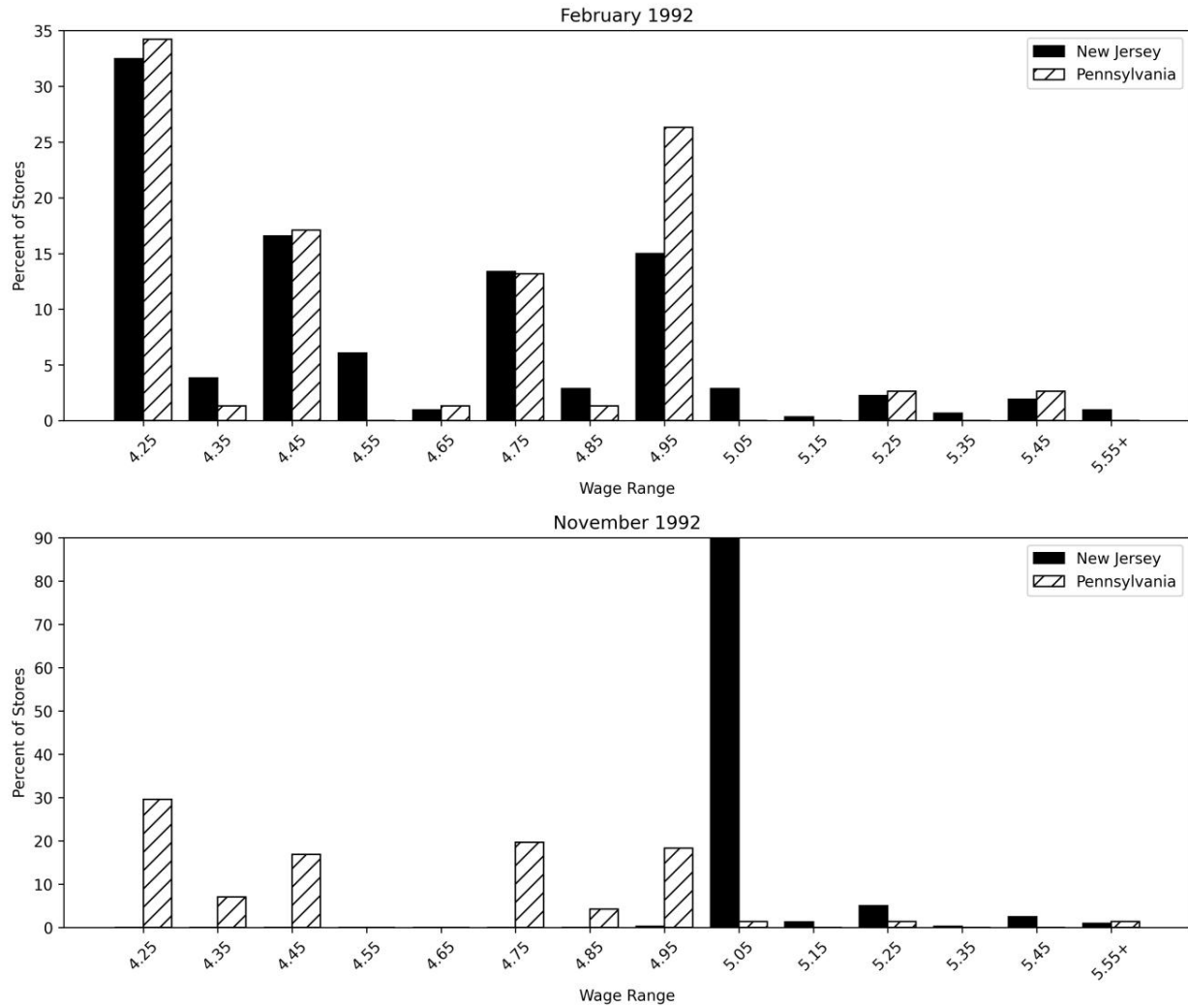
Model (v) is the one visible residual discrepancy in the main regression table. The workflow reports a GAP coefficient of 11.979237 and standard error 7.419121, compared with 11.91 and 7.39 in the paper. The recorded coefficient difference is 0.069237 and the standard-error difference is 0.029121. The standard error of regression is 8.752489 versus 8.75. These values are retained exactly as generated rather than altered to force equality. The separate author-check diagnostic in the workflow reproduces the 11.91 coefficient and 7.39 standard error, supporting the interpretation that the remaining difference is tied to an exact implementation/specification detail rather than to the broad sample construction.

The control diagnostics are also reported rather than omitted. The replication control p-values are 0.344251 in model (ii), 0.437360 in model (iv), and 0.404570 in model (v), compared with 0.34, 0.44, and 0.40 in the paper output. These are control-set diagnostics from the generated table and should not be misread as p-values for the main STATE or GAP coefficient.

4.4 Figure 1: distribution of starting wages

Figure 1 is regenerated directly from the final workflow. The underlying nonmissing wage counts are 314 New Jersey and 76 Pennsylvania observations in wave 1, and 318 New Jersey and 71 Pennsylvania observations in wave 2. The figure plots the percentage of observed stores at each wage range shown below.

Figure 1. Distribution of Starting Wage Rates



Replication of Figure 1. The plotted percentages are generated from the final replication workflow.

Figure 1 plotting values from outputs/tables/figure1_wage_distribution.csv

Wage range	Wave 1 NJ (%)	Wave 1 PA (%)	Wave 2 NJ (%)	Wave 2 PA (%)
4.25	32.484076	34.210526	0.000000	29.577465
4.35	3.821656	1.315789	0.000000	7.042254
4.45	16.560510	17.105263	0.000000	16.901408
4.55	6.050955	0.000000	0.000000	0.000000
4.65	0.955414	1.315789	0.000000	0.000000
4.75	13.375796	13.157895	0.000000	19.718310
4.85	2.866242	1.315789	0.000000	4.225352
4.95	14.968153	26.315789	0.314465	18.309859
5.05	2.866242	0.000000	89.622642	1.408451
5.15	0.318471	0.000000	1.257862	0.000000
5.25	2.229299	2.631579	5.031447	1.408451
5.35	0.636943	0.000000	0.314465	0.000000
5.45	1.910828	2.631579	2.515723	0.000000
5.55+	0.955414	0.000000	0.943396	1.408451

Before the policy change, the two distributions have substantial mass at the same low-wage points. At \$4.25, the wave-1 shares are 32.484076% in New Jersey and 34.210526% in Pennsylvania; at \$4.45 they are 16.560510% and 17.105263%; and at \$4.75 they are 13.375796% and 13.157895%. Pennsylvania has a larger wave-1 share at \$4.95 (26.315789%) than New Jersey (14.968153%).

After the reform, the New Jersey distribution changes sharply. The wave-2 New Jersey share at \$5.05 is 89.622642%, while the Pennsylvania share at the same wage is 1.408451%. Pennsylvania instead retains large shares at \$4.25 (29.577465%), \$4.45 (16.901408%), \$4.75 (19.718310%), and \$4.95 (18.309859%). The New Jersey wave-2 shares below \$4.95 are 0.000000% in every displayed wage range, with 0.314465% at \$4.95. This is the central visual pattern the replicated figure is intended to reproduce.

The figure-validation routine uses the same generated distribution and checks the New Jersey wave-2 \$5.05 value against the benchmark 89.62. The exact output value is 89.622642%; the validation summary displays 89.623 after its own reporting format. These are the same generated result shown at different reporting precision, not different statistics.

5. Independent Extension: Heterogeneous Effects Across Restaurant Chains

5.1 Motivation

The core replication estimates an average relative employment response. However, fast-food chains may differ in pricing, staffing, franchise structure, product mix or adjustment margins. A positive average effect could therefore conceal meaningful heterogeneity. The extension asks whether the New Jersey-Pennsylvania employment difference varies across Burger King, KFC, Roy Rogers and Wendy's.

The extension uses the balanced sample of 384 restaurants with observed FTE employment in both waves. Chain-specific regressions use heteroskedasticity-robust (HC1) standard errors. The Pennsylvania subsamples are relatively small for several chains, so the analysis is treated as an informative extension rather than a high-powered definitive test.

5.2 Extension specification

For each chain c , the chain-specific DiD is computed as

$$DiD_c = E[\Delta FTE_i | NJ_i = 1, Chain_i = c] - E[\Delta FTE_i | NJ_i = 0, Chain_i = c].$$

A pooled heterogeneity model is

$$\Delta FTE_i = \alpha + \beta NJ_i + \sum_c \gamma_c Chain_{ic} + \sum_c \delta_c (NJ_i \times Chain_{ic}) + \varepsilon_i,$$

with Burger King as the reference chain. The joint test is

$$H_0: \delta_{KFC} = \delta_{Roy\ Rogers} = \delta_{Wendy's} = 0.$$

Rejection would imply statistically detectable differences in the New Jersey employment response across chains.

5.3 Extension results

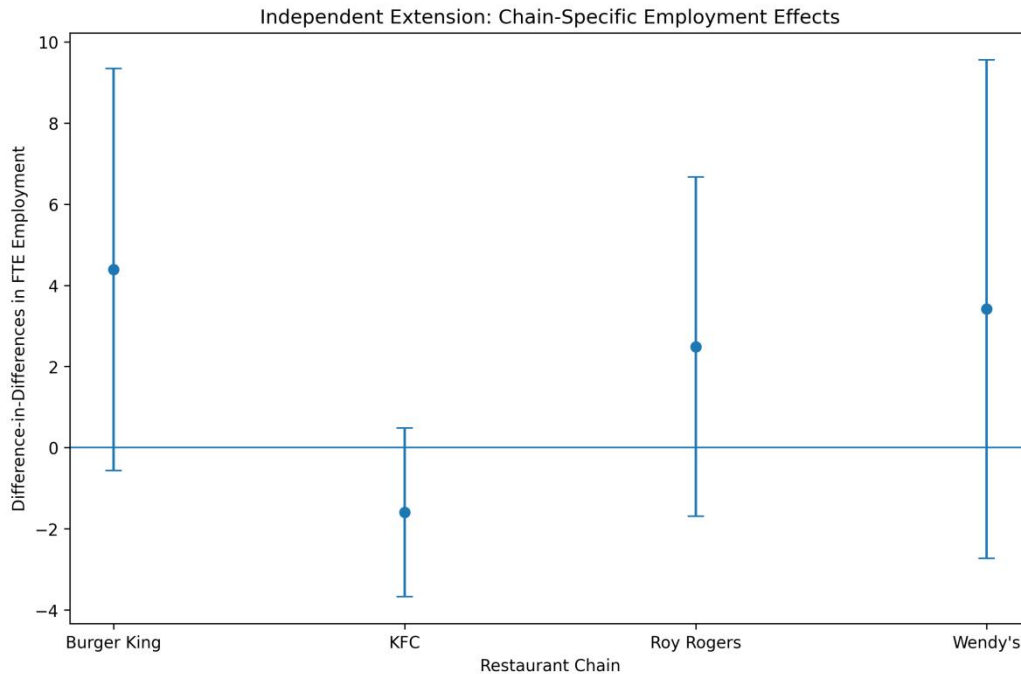
Table 4. Chain-specific employment responses

Chain	NJ N	PA N	NJ mean change	PA mean change	Chain DiD	Robust SE	p-value
Burger King	126	33	1.345	-3.045	4.390	2.529	0.083
KFC	67	12	0.698	2.292	-1.594	1.062	0.133
Roy Rogers	77	17	-1.438	-3.926	2.488	2.133	0.243
Wendy's	39	13	0.994	-2.423	3.417	3.134	0.276

Burger King has the largest positive point estimate, while KFC has a negative point estimate. None of the four chain-specific coefficients is individually significant at the 5% level. The joint heterogeneity test is

$$F = 2.602, \quad p = 0.0518.$$

Therefore, the null of equal chain-specific employment responses is **not rejected at the 5% level**. Because $p = 0.0518 < 0.10$, the result is nevertheless suggestive of heterogeneity at the 10% level. The distinction is important: the evidence is borderline, not conclusive.



Chain-specific DiD estimates with approximate 95% confidence intervals based on HC1 standard errors.

5.4 Economic interpretation of the extension

The extension adds information that is not visible in the average DiD estimate. Point estimates differ materially across chains, but the uncertainty is large. This is especially clear for Wendy's and Roy Rogers, where the Pennsylvania comparison groups contain only 13 and 17 restaurants. Hence wide confidence intervals are expected. The joint test is more informative than inspecting individual p-values alone, but it remains close to rather than below the 5% threshold.

The most defensible conclusion is therefore that **the data provide suggestive, but not conventionally significant, evidence that employment responses differed across restaurant chains**. A larger comparison sample or additional survey waves would be valuable for distinguishing true heterogeneity from sampling noise.

6. Validation, Replication Differences and Limitations

6.1 Automated validation

The project includes a separate validation script that compares core outputs with benchmark values using explicit tolerances. The final run reports:

Passed checks: 13/13

All core replication checks passed.

The validation includes Table 2 employment means, three Table 3 DiD quantities, five Table 4 coefficients and the Figure 1 wage-distribution check. Passing these tests is evidence that the implemented workflow reproduces the allocated outputs within transparent tolerances; it is not a claim that every coding choice is uniquely correct.

Table 5. Validation summary

Section	Test	Replication	Benchmark	Difference	Tolerance	Pass
Table 2	Wave 1 NJ FTE	20.439	20.44	-0.001	0.02	Yes
Table 2	Wave 1 PA FTE	23.331	23.33	0.001	0.02	Yes
Table 2	Wave 2 NJ FTE	21.027	21.03	-0.003	0.02	Yes
Table 2	Wave 2 PA FTE	21.166	21.17	-0.004	0.02	Yes
Table 3	Main DiD	2.754	2.76	-0.006	0.02	Yes
Table 3	Balanced-sample DiD	2.750	2.75	0.000	0.02	Yes
Table 3	Temporary-closure sensitivity	2.509	2.51	-0.001	0.02	Yes
Table 4	Model (i) NJ coefficient	2.326	2.33	-0.004	0.02	Yes
Table 4	Model (ii) NJ coefficient	2.304	2.30	0.004	0.02	Yes
Table 4	Model (iii) GAP coefficient	15.653	15.65	0.003	0.03	Yes
Table 4	Model (iv) GAP coefficient	14.916	14.92	-0.004	0.03	Yes
Table 4	Model (v) GAP coefficient	11.979	11.91	0.069	0.10	Yes
Figure 1	Wave 2 NJ share near \$5.05	89.623	89.62	0.003	0.05	Yes

6.2 Important replication differences

The project distinguishes harmless rounding differences from substantive specification differences. The main Table 3 DiD, for example, is 2.754 in the computation and 2.76 in the paper after rounding. That difference is mechanical. Table 4 model (v) is different: the replication obtains 11.979 while the paper reports 11.91. The discrepancy is small relative to the standard error and remains within the stated validation tolerance, but it is explicitly recorded. A separate diagnostic against author-supplied checking code reproduces the published 11.91/7.39 pair.

This treatment of discrepancies is preferable to altering inputs or coefficients by hand. The repository preserves the generated values, the benchmark values and the validation tolerances so an examiner can see exactly what was reproduced and where a small gap remains.

6.3 Limitations

Several limitations should constrain interpretation.

First, the design has only one pre-policy survey wave, so the parallel-trends assumption cannot be directly examined using pre-treatment dynamics. Second, restaurant employment is self-reported by survey respondents and therefore may contain measurement error. Third, the treatment is a state-level policy episode affecting one sector in one historical setting; external validity to other wage levels, sectors or periods is not automatic. Fourth, Table 3 and Table 4 use different sample restrictions, so small differences between simple and regression-adjusted estimates should be expected. Fifth, the chain heterogeneity extension has small Pennsylvania cell sizes for several chains, reducing statistical power.

Finally, the project is a replication exercise, not a claim to resolve the broader minimum-wage literature. Its main contribution is to show that the allocated Card-Krueger results can be reconstructed transparently from the supplied data, and to extend the same data with a clearly defined additional question.

7. Reproducibility and AI-Assisted Workflow

7.1 Reproducibility package

The complete project is organized so that the principal outputs can be regenerated from the raw input without hand-editing data or code. From the repository root, the intended sequence is:

```
pip install -r requirements.txt
python code/run_all.py
```

The master script executes raw-data inspection, cleaning, Tables 2-4, Figure 1, validation and the independent extension in sequence. Generated files are written to `data/processed/`, `outputs/tables/` and `outputs/figures/`. The final workflow uses only relative paths.

Fresh-clone reproducibility check. After the required raw input is placed at `data/raw/public.csv`, running the two commands above regenerates the processed dataset, replication tables, validation summary, and figures from the submitted inputs. The workflow does not require manual editing of the data, hand-entering of reported coefficients, or machine-specific absolute paths.

Repository documentation. `README.md` provides setup and execution instructions; `skills/SKILL.md` documents the reusable replication workflow; `requirements.txt` records the Python dependencies; and `AI_USE_DISCLOSURE.md` records where AI assistance was used and how it was checked. The Git history records meaningful stages of development rather than only a single final upload.

Project repository:

https://github.com/WUYi1128/WU-Yi_3036760890

7.2 AI use and verification

Artificial-intelligence tools were used to assist with project planning, Python code structure, debugging, documentation and interpretation. AI output was not treated as authoritative. Code was executed locally; results were compared with the original paper and benchmark values; the final validation script checks 13 core quantities; and suggestions involving incorrect paths, unsupported claims or over-strong statistical interpretation were corrected or rejected. Concrete examples include replacing machine-specific absolute paths with repository-relative paths and rejecting the claim that the chain-heterogeneity extension is statistically significant at the 5% level when its joint-test p-value is 0.0518. All final data choices, empirical specifications, validation decisions, interpretations and conclusions were reviewed by the student. A detailed record is provided separately in `AI_USE_DISCLOSURE.md`, as required by the assignment.

8. Conclusion

This project successfully reconstructs the main empirical outputs assigned from Card and Krueger (1994). The descriptive statistics reproduce the pre- and post-policy employment and wage patterns, the central DiD estimate is 2.754 compared with the published 2.76, and the regression-adjusted models match the published coefficients closely. Figure 1 also reproduces the sharp post-reform concentration of New Jersey starting wages around the new \$5.05 minimum. All 13 automated validation checks pass.

The replication therefore supports the paper's empirical finding for this sample: employment in New Jersey did not decline relative to eastern Pennsylvania after the minimum-wage increase; the estimated relative change is positive. The report does not interpret this single episode as a universal minimum-wage effect, because the design relies on the validity of the Pennsylvania comparison group and contains only one pre-treatment wave.

The independent chain-level extension finds economically different point estimates across restaurant chains, but the joint heterogeneity test yields $p = 0.0518$. The evidence is thus suggestive at 10% but does not meet the

conventional 5% threshold. This extension illustrates the value of moving beyond a headline average while also demonstrating why statistical uncertainty and subgroup sample size must be reported carefully.

References

Card, D., & Krueger, A. B. (1994). Minimum wages and employment: A case study of the fast-food industry in New Jersey and Pennsylvania. *American Economic Review*, 84(4), 772-793.

Card, D. (n.d.). Available data sets: Minimum Wages and Employment. David Card, University of California, Berkeley. Replication materials associated with Card and Krueger (1994).

Appendix A. Output Map

The final workflow generates the following principal files:

- data/processed/analysis_sample.csv
- outputs/tables/table2_replication.csv
- outputs/tables/table3_replication.csv
- outputs/tables/table4_replication.csv
- outputs/tables/figure1_wage_distribution.csv
- outputs/tables/validation_summary.csv
- outputs/tables/extension_chain_did.csv
- outputs/tables/extension_chain_heterogeneity_test.csv
- outputs/figures/figure1_wage_distribution.png
- outputs/figures/extension_chain_did.png

README.md (setup and reproduction instructions)

skills/SKILL.md (reusable workflow documentation)

AI_USE_DISCLOSURE.md (AI-use and verification record)

The code sequence is:

1. 01_inspect_data.py
2. 02_clean_data.py
3. 03_replicate_table2.py
4. 04_replicate_table3.py
5. 05_replicate_table4.py
6. 06_replicate_figure1.py
7. 07_validation.py
8. 08_extension.py
9. run_all.py orchestrates the complete workflow.